

DB Lab Task

Project Name:

Air-Line Management System

Submitted By:

Nawaal Farakh 70154162

Dua Sohail 70154366

Task 1 Description & Scenario

Project Description

NexusAir Crisis Hub is an advanced airline operations management system specially designed to manage emergency situations and operational disruptions in airlines.

Unlike a normal airline booking system, this system focuses on real-life airline crises such as:

- Flight delays
- Pilot illness
- Aircraft replacement
- Lost or misdirected baggage
- Passenger rebooking
- Operational alerts

The system helps airline staff monitor and resolve problems quickly while keeping passengers informed.

Scenario:

NexusAir Airlines operates international and domestic flights daily. During operations, many unexpected situations occur.

For example:

- A pilot suddenly becomes sick before departure.
- A flight gets delayed because of weather conditions.
- An aircraft develops technical issues and must be replaced.
- Passenger baggage is sent to the wrong airport.
- Passengers miss connecting flights and need rebooking.

The NexusAir Crisis Hub system stores all disruptions in the database and allows airline staff to track, manage, and resolve these situations efficiently.

The system also sends alerts to operational staff and keeps records of all actions taken during the crisis.

Features of the System

1. Flight Operations Management

- Manage flights
- Monitor delays and schedules
- Track flight status

2. Aircraft Management

- Store aircraft information
- Assign aircraft to flights
- Replace aircraft during emergencies

3. Pilot & Staff Management

- Manage pilots and airline staff
- Handle pilot assignments
- Track staff roles

4. Crisis & Disruption Management

- Record operational disruptions
- Store delay reasons
- Handle pilot illness cases
- Record aircraft swaps

5. Baggage Management

- Track baggage location
- Identify misdirected baggage
- Resolve baggage issues

6. Passenger Booking & Rebooking

- Store booking records
- Rebook passengers during delays/cancellations

7. Alert Management

- Generate alerts for airline staff
- Notify operations team during emergencies

Outcome of the Project

The system will:

- Improve airline operational efficiency
- Help staff respond quickly during emergencies
- Reduce passenger inconvenience
- Improve baggage tracking
- Maintain proper airline records

- Demonstrate advanced ER/EER database concepts

Task 2 Entities, Attributes, and Relationships

Entities & Attributes

1. Flight

Attribute	Description
FlightID (PK)	Unique Flight ID
FlightNo	Flight Number
Origin	Departure city
Destination	Arrival city
DepartureTime	Flight departure
ArrivalTime	Flight arrival
Status	Flight status

2. Aircraft

Attribute	Description
AircraftID (PK)	Aircraft ID
AircraftType	Aircraft model
Capacity	Passenger capacity
Status	Active/Inactive

3. Pilot

Attribute	Description
PilotID (PK)	Pilot ID
PilotName	Pilot name
LicenseNo	License number
Phone	Contact
Status	Available/Sick

4. Flight_Assignment

Attribute	Description
AssignmentID (PK)	Assignment ID
FlightID (FK)	Linked flight
PilotID (FK)	Linked pilot
Role	Captain/Co-Pilot

5. Passenger

Attribute	Description
PassengerID (PK)	Passenger ID
FullName	Passenger name
Phone	Contact
Email	Email

6. Booking

Attribute	Description
BookingID (PK)	Booking ID
PassengerID (FK)	Linked passenger
FlightID (FK)	Linked flight
BookingDate	Date of booking
Status	Confirmed/Cancelled

7. Rebooking

Attribute	Description
RebookingID (PK)	Rebooking ID
BookingID (FK)	Original booking
NewFlightID (FK)	New assigned flight
Reason	Delay/Cancellation
RebookedAt	Rebooking date

8. Disruption

Attribute	Description
DisruptionID (PK)	Disruption ID
FlightID (FK)	Affected flight
Type	Delay/Pilot Sick/etc
Description	Details
ReportedAt	Time reported
Status	Open/Resolved

9. Disruption_Resolution

Attribute	Description
ResolutionID (PK)	Resolution ID
DisruptionID (FK)	Linked disruption
ActionTaken	Resolution details
ResolvedAt	Resolution time
ResolvedBy	Staff ID

10. Baggage

Attribute	Description
BaggageID (PK)	Baggage ID
PassengerID (FK)	Owner
TagNo	Baggage tag
CheckInFlightID	Checked flight
Status	CheckedIn/InTransit/Misdirected

11. Baggage_Misdirect

Attribute	Description
MisdirectID (PK)	Misdirect ID

Attribute	Description
BaggageID (FK)	Linked baggage
WrongDestination	Wrong airport
FoundAt	Current location
Status	Open/Resolved

12. Baggage_Resolution

Attribute	Description
ResolutionID (PK)	Resolution ID
MisdirectID (FK)	Linked issue
ActionTaken	Courier/Transfer
DeliveredAt	Delivery date
DeliveredBy	Staff member

13. Staff

Attribute	Description
StaffID (PK)	Staff ID
StaffName	Staff name
Role	Operations/Admin
Phone	Contact

14. Alert

Attribute	Description
AlertID (PK)	Alert ID
StaffID (FK)	Staff receiving alert
Message	Alert message
CreatedAt	Alert time
IsRead	Yes/No

Relationships:

Relationship	Type
Aircraft operates Flight	One-to-Many
Pilot assigned to Flight	Many-to-Many
Passenger makes Booking	One-to-Many
Booking may create Rebooking	One-to-Many
Flight has Disruption	One-to-Many
Disruption has Resolution	One-to-Many
Passenger owns Baggage	One-to-Many
Baggage has Misdirect Case	One-to-Many
Staff generates Alert	One-to-Many